Unit 6, Lesson 16

# Base-Ten Diagrams to Represent Division

## Standards

| | |
|---|---|
| Addressing | **4.NBT.B.6** |
| Building Toward | **4.NBT.B.6** |

## Instructional Routines

- Which Three Go Together?

## Goals

- Interpret (orally) methods for division based on place-value.
- Use base-ten blocks or diagrams to divide two- or three-digit numbers by one-digit divisors.

## Student Facing Learning Goals

Let's make sense of base-ten diagrams that represent division.

## Lesson Purpose

The purpose of this lesson is for students to find the quotients of two-digit and three-digit dividends and one-digit divisors as they consider base-ten diagrams that represent the problems.

## Narrative

In the previous lesson, students used base-ten blocks to represent division and represented their work on paper. This lesson focuses on base-ten diagrams to represent dividing by decomposing the dividend by place value—decomposing a larger unit to 10 of a smaller unit—and by reasoning in terms of equal-size groups. Students recall that they can exchange or decompose one or more units of a higher place value for 10 units of the next lower place value in order to have enough units to put into equal groups. This lesson sets the groundwork for the reasoning students use in a future course when they learn to divide multi-digit numbers using the standard division algorithm, which relies on dividing by place value, one digit at a time.

## Access for Students with Disabilities

- Engagement

## Access for English Learners

- MLR8

## Required Materials

**Materials to Gather**

- Base-ten blocks: Activity 1, Activity 2

## Lesson Timeline

| | |
|---|---|
| Warm-up | 10 min |
| Activity 1 | 15 min |
| Activity 2 | 20 min |

## Teacher Reflection Questions

How did the representations in today's lesson support students in dividing multi-digit numbers?

| | |
|---|---|
| Synthesis Estimate | 10 min |
| Cool-down | 5 min |

# Warm-up

10 min

Which Three Go Together: Base-Ten Diagrams

## Standards

Building Toward **4.NBT.B.6**

## Instructional Routines

- Which Three Go Together?

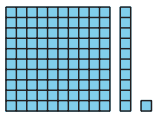
This *Warm-up* prompts students to carefully analyze and compare features of base-ten diagrams, looking not only at the number and types of shapes in each diagram, but also the value each diagram represents. Students have a reason to use language precisely (MP6) as they recall what they know about representations of numbers in base-ten. The activity also enables the teacher to hear how students talk about these representations.

The analysis prepares students for the activities in the lesson, in which they use base-ten diagrams to find whole-number quotients.

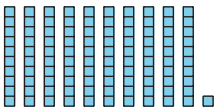
## Student Task Statement

Which 3 go together?

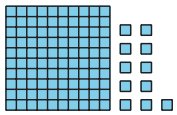
A



B



C



D



## Student Response

Sample responses:

- A, B, and C go together because they have a column with just a single square.
- A, B, and D go together because they have tens.
- A, C, and D go together because they show 111.
- B, C, and D go together because they have 10 blocks.

## Launch

- Groups of 2
- Display image.
- "Pick 3 that go together. Be ready to share why they go together."
- 1 minute: quiet think time

## Activity

- "Discuss your thinking with your partner."
- 2–3 minutes: partner discussion
- Record responses.

## Activity Synthesis

- "How is it that A and C both show 111?" (If a small square represents 1, then a rectangle is 10 and a large square is 100. In A: $100 + 10 + 1 = 111$. In C: $100 + 11 = 111$.)
- "How do we know that D also shows 111?" (Each group in D represents $(3 \times 10) + 7$ or 37. Three groups of 37 makes 111.)
- "Suppose we don't know what a small square represents except that it represents the same value in all diagrams. Can we tell if C and D represent the same value? How?" (Yes. We know that 10 small

squares make 1 rectangle and 10 rectangles make 1 large square. In D, we'd have 21 small squares and 9 rectangles. Trading 10 small squares for a rectangle gives 10 rectangles and 11 small squares, which is equal to 1 large square and 11 small squares.)

# Activity 1

15 min

Divide with Diagrams or Blocks

## Standards

Addressing **4.NBT.B.6**

The purpose of this activity is for students to use base-ten diagrams to find quotients of two-digit dividends and single-digit divisors. They think about distributing the pieces in the diagram into equal-size groups, decomposing a higher-value piece with 10 of the lower-value pieces as needed to divide.

## Access for English Language Learners

*MLR8 Discussion Supports.* Use multimodal examples to show the meaning of place value. Use verbal descriptions along with gestures, drawings, or concrete objects to show how a base-ten block is equivalent to 10 ones blocks and how they are interchangeable.
*Advances: Listening, Representing*

## Required Materials

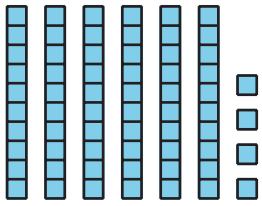
**Materials to Gather**

- Base-ten blocks: Activity 1

## Student Task Statement

1. Priya draws a base-ten diagram to find the value of $64 \div 4$. A rectangle represents 10. A small square represents 1.

   Use the diagram (or actual blocks) to help Priya complete the division. Explain or show your reasoning.



2. Use this base-ten diagram (or actual blocks) to

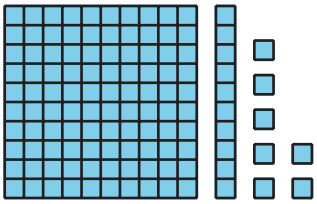
## Launch

- Groups of 4
- Give students access to base-ten blocks.
- Display the first diagram. Make sure students can explain why it represents 64.

## Activity

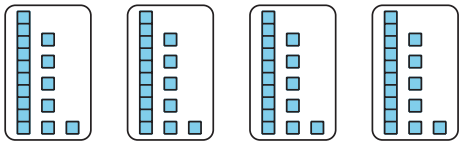
- 5 minutes: independent work time
- 2 minutes: group discussion
- Monitor for students who see that a larger piece can be decomposed into 10 of the next smaller piece to help with distribution.

find the value of $117 \div 3$.

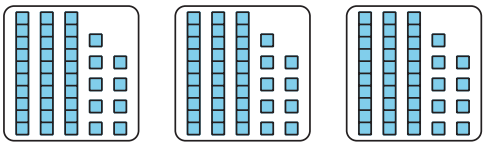


## Student Response

1. 16. Sample response: Four of the 6 tens can be put into 4 groups of 1 ten. The other 2 tens can be decomposed into 20 ones. There are now 24 ones, which can be divided into 4 groups of 6. One ten and 6 ones make 16.



2. 39. Sample response: One hundred can be decomposed into 10 tens. Nine of the 11 tens can be put into 3 groups of 3 tens. There are now 2 tens which can be composed into 20 ones. There are now 27 ones, which can be divided into 3 groups of 9. Three tens and 9 ones make 39.



## Activity Synthesis

- Invite students to share their responses and reasoning.
- Make sure students see that:
  - We can think of $64 \div 4$ as putting 6 tens and 4 ones into 4 equal groups, and $117 \div 3$ as putting 1 hundred 1 ten and 7 ones into 3 equal groups.
  - To divide the base-ten pieces, we can decompose a piece representing a larger place value with 10 of the next smaller place value.

# Activity 2

20 min

Help Noah Get Unstuck

## Standards

Addressing **4.NBT.B.6**

The purpose of this activity is for students to continue to use base-ten representations and to reason about equal-size groups to find whole-number quotients. The work reinforces the idea of decomposing a hundred into 10 tens as needed to perform division.

Students explicitly use place value understanding to decompose hundreds and tens (MP7) while making sense of a students' reason to help him complete the division problem (MP3).

## Access for Students with Disabilities

*Engagement: Provide Access by Recruiting Interest.* Optimize meaning and value. Invite students to share examples from their own lives in which they might need to divide three-digit numbers by one-digit numbers. Ask them to imagine and share why Noah might be dividing 235 by 5.
*Supports accessibility for: Attention, Social-Emotional Functioning*

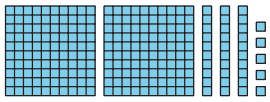
## Required Materials

**Materials to Gather**

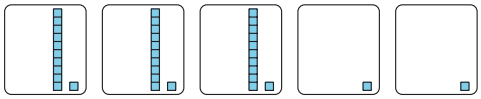
- Base-ten blocks: Activity 2

## Student Task Statement

1. This diagram represents 235.



Noah draws this diagram to find $235 \div 5$ and then gets stuck.



He says, "There are not enough of the hundreds or the tens pieces to put into 5 groups."

Explain or show how Noah could find $235 \div 5$ with his diagram.

2. Find the value of $432 \div 6$. Show your thinking using diagrams, symbols, or other representations. Use base-ten diagrams or blocks if you find them helpful.

## Student Response

1. Sample response: Noah can exchange the 2 hundreds for 20 tens, which can make 5 groups of 4 tens. The other 3 tens can be exchanged for 30 ones. There are now 35 ones in total, which can make 5 groups of 7 ones with no remainder. Each group has 4 tens and 7 ones, which is 47.
2. Sample response: 432 can be represented by 4 hundreds, 3 tens, and 2 ones.
    - Exchange 4 hundreds for 40 tens. Now, there are 43 tens.
    - Use 42 tens to make 6 groups of 7 tens. There is 1 ten left over.

## Launch

- Groups of 2
- Give students access to base-ten blocks.
- Display the first diagram.
- "How does the diagram represent 235?" (A large square represents 100. A rectangle represents 10. A small square represents 1.)

## Activity

- 4–5 minutes: independent work time on the first question
- Monitor for students who see the 2 hundreds as 20 tens and those who see them as 200 ones.
- Pause after the first question. Make sure students see that the 2 hundreds can be decomposed into 20 tens (or 200 ones) and split into 5 equal groups, and the 3 tens can be decomposed into 30 ones and split into 5 groups. Complete the second diagram to illustrate this reasoning.
- 5 minutes: partner work time on the last question

## Activity Synthesis

- Select students to share their responses and reasoning for the last question. Display for all to see.
- Highlight the idea that each unit can be decomposed into 10 units of a lower place value to make it possible to create equal-sized groups.

- Exchange the 1 ten for 10 ones. There are now 12 ones.
- Put the 12 ones into 6 groups of 2 ones.
- Each group has 7 tens and 2 ones, or 72.

## Lesson Synthesis

"Today we used base-ten diagrams and blocks to find quotients such as $712 \div 4$. Here is a student's unfinished work for finding $712 \div 4$. How would you complete it?"

Display:

"$712 \div 4$ means putting 7 hundreds + 1 ten + 2 ones into 4 equal groups."

| | | | |
|---|---|---|---|
| 1 hundred | 1 hundred | 1 hundred | 1 hundred |

Invite student volunteers to share their reasoning. (178. After putting 1 hundred in each group, there are 3 hundreds, 1 ten, and 2 ones left. The hundreds can be decomposed into tens and the tens can be decomposed into ones so that there's enough to put into 4 groups.

$$\begin{aligned} &3 \text{ hundreds} + 1 \text{ ten} + 2 \text{ ones} \\ &= 30 \text{ tens} + 1 \text{ ten} + 2 \text{ ones} \\ &= 28 \text{ tens} + 3 \text{ tens} + 2 \text{ ones} \\ &= 28 \text{ tens} + 30 \text{ ones} + 2 \text{ ones} \\ &= 28 \text{ tens} + 32 \text{ ones} \end{aligned}$$

$28 \div 4 = 7$, so 7 tens in each group.

$32 \div 4 = 8$, so 8 ones in each group.)

| | | | |
|---|---|---|---|
| 1 hundred<br>7 tens<br>8 ones | 1 hundred<br>7 tens<br>8 ones | 1 hundred<br>7 tens<br>8 ones | 1 hundred<br>7 tens<br>8 ones |

### Suggested Centers

- Compare (1–5), Stage 4: Divide within 100 with One-Digit Divisors (Supporting)
- Rolling for Fractions (3–5), Stage 2: Multiply a Fraction by a Whole Number (Supporting)

## Cool-down

5 min

Find the Value of a Quotient

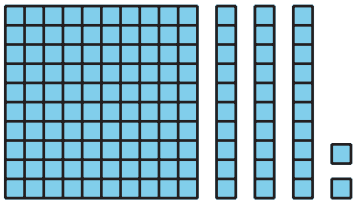
### Standards

Addressing **4.NBT.B.6**

## Student Task Statement

Find the value of $132 \div 4$. Show your thinking using diagrams, symbols, or other representations.

The base-ten diagram represents 132. Use the diagram if you find it helpful.



## Student Response

33. Sample responses:

- I know that $132 = 100 + 32$. I also know that $100 \div 4 = 25$ and $32 \div 4 = 8$, so $132 \div 4$ is the sum of 25 and 8, which is 33.
- The large square represents 1 hundred and can be decomposed into 10 tens. Now, we have 13 tens. Twelve of the tens can be put into 4 groups of 3 tens. The last ten can be decomposed into 10 ones. There are now 12 ones, or 4 groups of 3 ones. Three tens and 3 ones is 33.
- I know that $120 \div 4 = 30$ and $12 \div 4 = 3$, so $132 \div 4 = 33$.

## Responding to Student Thinking

The work in this lesson builds on the expanded form of numbers developed in a prior unit.

Prior-Unit Support
Grade 2, Unit 5, Section A The Values of Three Digits

Students decompose the dividend incorrectly, resulting in errors.

Next Day Supports
During the *Activity Syntheses*, connect diagrams to expressions or equations.